

Rohith Mukku

Email | LinkedIn | Website | Google Scholar

EDUCATION

MS, Computer Science, New York University, GPA: 3.97/4.00

Jan 2021 – Dec 2022

BTech, Computer Science, Indian Institute of Technology Kanpur

Jul 2014 – Jun 2018

TECHNICAL SKILLS

Software Engineering

C/C++ • Python • Bash • AWS Lambda, AWS Fargate • Git • familiar with Java, Haskell, SQL

Machine Learning

PyTorch • scikit-learn • Pandas • vLLM • HuggingFace • Weights & Biases • MLFlow • AzureML • SLURM

RESEARCH EXPERIENCE

- *The Surprising Effectiveness of Representation Learning for Visual Imitation*. RSS 2022. (Acknowledged contributor — code & ablation experiments)
- *No Offense Taken: Eliciting Offensiveness from Language Models*. arXiv:2310.00892, 2023. (5 citations - equal contribution)
- *Large Language Models for Page Stream Segmentation*. arXiv:2408.11981, 2024. (3 citations - co-author)
- *What We Learned From Deploying Fine-Tuned LLMs in Production*. Roots AI Engineering Blog, 2024.

RELEVANT EXPERIENCE

Meta

Sep 2024 – Present

Machine Learning Engineer, Voice AI Assistant Team, Reality Labs

New York

- Investigated **LLM-as-Judge** methodology — identified **self-preference bias** and developed calibration framework achieving **80%+** precision/recall vs human annotations (hard mismatch below **15%**)
- Owned evaluation framework for Product Help domain in **Meta AI 2.0**, assessing response quality using grounded evaluation across three retrieval contexts (internal KB, Google Search, hybrid)
- Investigated **RAG** quality improvement through embedding model selection and knowledge curation — systematic ablations achieved **+2% human task success rate** and **~15% P90 latency reduction** across international locales in **Meta AI 1.0**
- Engineered language-independent evaluation framework enabling **18-language voice AI expansion** with **100% international coverage** across **30+ languages**
- Curated instruction-following evaluation datasets to systematically identify model failure modes and guide post-training
- Investigated **Live Speech Translation** latency reduction via fine-tuning on incomplete utterances to eliminate partial/finalize heuristics; stop-token and text-delay approaches revealed **spBLEU improvement** with **WER tradeoff**
- Implemented accuracy parity testing framework for **E2E speech translation** on privacy-preserving secure compute for Smart Glasses

Roots AI

Feb 2023 – Sep 2024

Machine Learning Engineer

New York

- Fine-tuned **Mistral 7B** and **Phi-3** for insurance entity extraction; deployed quantized models via **vLLM** on Azure ML, achieving **25×** speedup over HuggingFace inference, processing **20–30M documents/year** at \$30K/year; built automated retraining pipeline
- Trained multimodal LLMs for insurance documents; fine-tuned models **outperformed GPT-4** on domain tasks
- Implemented text classification (**XGBoost/BERT**) and entity extraction pipelines; maintained **MLOps** infrastructure on **Azure ML Studio**
- Built and maintained **InsurGPT**, an LLM-powered document extraction product deployed on **Azure**

Amazon Web Services

May 2022 – Aug 2022

Software Development Engineer Intern

Seattle

- Migrated local **AWS EC2** component to serverless architecture using **AWS Fargate**, **CDK**, and **DynamoDB**

Samsung Research Institute Delhi

Jul 2018 – Dec 2020

Software Engineer

Noida

- Developed **C/C++** kernel modules and device drivers for **Tizen TV OS** (Samsung Smart TV platform)
- Developed app recommendation system using **contextual bandits** and Naive Bayes

PROJECTS

Out-Of-Distribution Detection using Normalizing Flows [Github] [Report]

Prof. Rajesh Ranganath

- Implemented **FlowGMM**, a **Normalizing Flow** model with **Gaussian Mixture Models** to detect OOD data in PyTorch; improved AUROC/AUPRC on MNIST, FashionMNIST, SVHN

Red Teaming Language Models [Github] [Report]

Prof. Sam Bowman

- Applied Zero-Shot, Few-Shot, Supervised, and RL methods for adversarial test case generation targeting **GPT-2 Large** and **BlenderBot**; replicated paper results

Self-Supervised Learning Competition [Final Repo] [Github] [Report]

Prof. LeCun, Prof. Canziani

- Applied SOTA self-supervised methods (**SimCLR**, **MoCo**) + active learning on **512k** unlabeled, **50k** labeled images; ranked **7th** of 23 teams with 36.7% test accuracy

Deep RL Implementations [Github]

- Implemented PPO, A2C, DQN, DDPG, VPG from scratch in PyTorch; trained agents on MuJoCo continuous control environments including Humanoid-v2

ACADEMIC POSITIONS

- **Research Assistant**, Prof. Lerrel Pinto, NYU (Sep 2021 – May 2022) — code & ablation experiments for **VINN** (SSL + kNN for visual imitation learning) [RSS 2022]; studied environment generalization [Blog]
- **Teaching Assistant**, Deep Learning course taught by Prof. LeCun, Prof. Bruna, Prof. Canziani (Fall 2021 & Fall 2022)